

Mainul Islam

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Education

B.Sc. in Electrical and Electronic Engineering (EEE)

Islamic University of Technology (IUT), Gazipur, Bangladesh

Jan 2020 – Jun 2024

CGPA: 3.87 / 4.00

Thesis — MonoSIM: Developed an open-source Software-in-the-Loop (SIL) framework for autonomous Ackermann vehicles using a monocular-vision, sliding-window lane-detection pipeline, and implemented and benchmarked Model Predictive Control (MPC) and PID strategies for path tracking within the simulation environment (published and presented).

Professional Experience

Research and Development Engineer / Embedded Systems Engineer — [Cybernetics Hi-Tech Solutions Pvt. Ltd.](#) Oct 2024 – Present

- **Autonomous Ground Vehicle (AGV)**: Built a sensor-fusion system combining cameras, wheel odometry, and visual markers for accurate self-localization. Added LiDAR-based obstacle avoidance and unified navigation, localization, and motion control into a single ROS2 system. Resolved a network bottleneck limiting inter-robot communication, and engineered unattended system bring-up so the full autonomy stack self-initializes and comes online automatically at power-up.
- **Man-Portable Teleoperated Robot**: Built the communication and control system for an 11-motor robot, including safety interlocks and a fault-monitoring suite with automated fault handling and recovery. Added a redundant wireless / fiber-optic link so the robot stays controllable if the primary connection drops. Designed the manipulator-arm motion control and a low-latency live-video system with 5 cameras, a 3D digital-twin viewer, and remote camera control.
- **Toll-Booth Automation**: Built an automated system to detect vehicles, verify payment, and flag potential theft, using vehicle sensors and license-plate recognition. Designed a custom control PCB with tamper protection.

Part-Time Lecturer, Department of EEE — [Ahsanullah University of Science and Technology \(AUST\)](#) Jul 2025 – Jun 2026

- Instructed the **C++ Programming (EEE-1110)** and **Digital Signal Processing (EEE-3218)** laboratories across two consecutive semesters, including lecturing, lab-manual preparation, and student assessment.

Publications

Peer-Reviewed Journal Articles

- A. U. R. Adib, **M. Islam**, M. S. Abid, R. Ahshan, “A deep learning based framework for solar panel segmentation and fault classification enhanced with explainable AI,” *Solar Energy*, vol. 302, 2025. (Impact Factor: 6.6)
DOI: [10.1016/j.solener.2025.114058](https://doi.org/10.1016/j.solener.2025.114058)

Peer-Reviewed Conference Papers

- S. Rahman, N. Hasin, **M. Islam**, M. Z. A. Rony, G. Sarowar, “MonoSIM: An open-source SIL framework for Ackermann vehicular systems with monocular vision,” *IEEE 12th Intl. Conf. on Automation, Robotics and Applications (ICARA)*, 2026.
DOI: [10.1109/ICARA69401.2026.11480327](https://doi.org/10.1109/ICARA69401.2026.11480327)
- N. Hasin, M. A. R. Nishat, **M. Islam**, K. S. A. Hasan, A. Newaz, “A robust deep learning framework for Bangla license plate recognition using YOLO and vision-language OCR,” *IEEE Intl. Conf. on AI and Data Analytics (ICAD)*, 2026.
DOI: [10.1109/ICAD69378.2026.11608728](https://doi.org/10.1109/ICAD69378.2026.11608728)
- H. Haque, **M. Islam**, M. A. I. Alvy, M. S. Islam, M. A. Razzak, “Long-term energy demand analysis using machine learning algorithms: A case study in Bangladesh,” *IEEE ICECIE*, 2024.
DOI: [10.1109/ICECIE63774.2024.10815643](https://doi.org/10.1109/ICECIE63774.2024.10815643)
- M. S. Tanvir, F. A. Tonmoy, M. M. Islam, **M. Islam**, M. Hasan, “Optimizing master-slave broadcast and unicast communication in multi-node STM32 networks using UART and RS485 protocols,” *IEEE ICCIT*, 2024.
DOI: [10.1109/ICCIT64611.2024.11022430](https://doi.org/10.1109/ICCIT64611.2024.11022430)

Manuscripts Under Review

- S. Rahman, M. K. Hossain, A. Banik, **M. Islam**, A. Al-Hysam, I. Ahmed, “Topological survey of Zeta power converters from classical designs to emerging architectures.”
- M. M. Islam, **M. Islam**, M. S. Tanvir, “Speed synchronization of a dual-BLDC tracked differential-drive vehicle using a GA-tuned multi-rate cross-coupled controller.”

Projects

- **Man-Transportable EOD ROV (JS2.0)**: CANopen control of 11 servos over dual CAN, STM32 SBUS-over-UDP link with wireless failover, closed-form 7-DOF inverse kinematics, layered fail-safe logic, and the Android operator console (*in progress*).
- **Jontro Soinik 1.0 & Smart Soinik 1.0**: Addressable-UART modular control, sensor and power-system integration up to PCB fabrication, and field testing.
- **Autonomous Guided Vehicle — CYBERFLEET**: Built the ROS2 autonomy stack — 5-state EKF (AprilTag + odometry), space-time A* planner, trajectory-tracking controller, and the fleet coordinator.
- **Autonomous Target-Tracking Drone**: Onboard-vision target tracking (YOLO on Raspberry Pi 5; DeepSORT/KCF), dataset creation, and flight-time optimization.
- **Surveillance Drone (SD15-N)**: Flight-time analysis and optimization (drag, thrust, battery) and field testing.
- **Toll-Booth Automation**: STM32 based DMA-UART + WIZnet W5500 TCP stack, and three custom PCBs design, manufacture and assembly.
- **Offline Hardware CryptoWallet**: System state machine, display/touch UI and QR-based offline transaction signing.
- **8-bit SAP-1 Computer**: Designed an educational 8-bit CPU (ALU, control unit, registers) with reduced T-states executing the fetch–decode–execute cycle.
- **LiPo Battery Management System (BMS)**: MATLAB cell-protection simulation (load cut-off outside 3.7–4.2 V, 2S–6S) and PCB design.
- **Pulse-Width Measurement & Generation (AT89S52)**: 8051 assembly firmware for frequency/duty-cycle measurement with threshold alarm and summed-frequency signal generation.
- **IUT Mars Rover — Project Altair**: Electrical system design — PCB design, hardware integration, and debugging.
- **IUT Mars Rover — Team Avijatrik**: Designed the wheel-drive board and several PCBs/sub-circuits; wiring and debugging.

Technical Skills

Programming: Python, C/C++, MATLAB, Embedded C

ML / DL: PyTorch, TensorFlow, scikit-learn, YOLO, Explainable AI (LIME/SHAP), Vision-Language OCR

Robotics / Control: ROS2, Extended Kalman Filter (EKF), sensor fusion, inverse kinematics, MPC, PID, LiDAR/vision perception, AprilTag/vision-based localization

Embedded & Communication: CAN bus, RS-485, UART, Modbus, STM32, PCB design, real-time distributed control

CAD / 3D Design: Autodesk Fusion, Onshape

Honors & Achievements

- **IRC 2024 — 6th Global**: International Rover Challenge; ABEX Best Science Team (Project Altair, 1st in Bangladesh).
- **ERC 2023 — 16th Global**: European Rover Challenge (Project Altair).
- **IRDC 2021 — 11th Global**: Indian Rover Design Challenge (Team Avijatrik, 1st in Bangladesh).
- **MATLAB Cody**: Global rank 117; 914 problems solved; 26 badges.

References

Prof. Dr. Golam Sarowar — B.Sc. Thesis Supervisor

Professor, Department of EEE, Islamic University of Technology (IUT)

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